

Here is a clean, structured reference documentation template based on the steps you just performed. You can save this as a README.md or keep it for future reference when you need to re-flash the module.

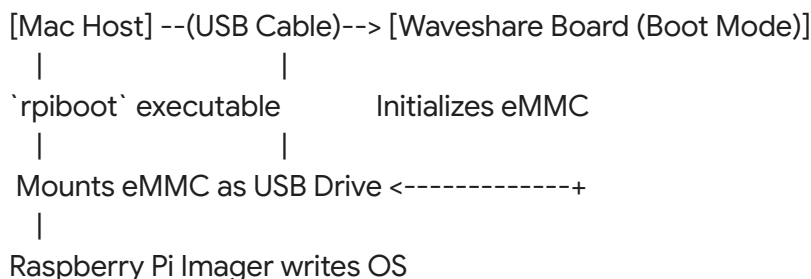
Raspberry Pi CM4 eMMC Flashing Guide (macOS)

This document provides the standard operating procedure for flashing a Raspberry Pi Compute Module 4 (CM4) with onboard eMMC storage using a Waveshare CM4 Expansion Board and a macOS host machine.

Prerequisites

- **Hardware:**
 - Raspberry Pi CM4 with eMMC storage.
 - Compatible Waveshare CM4 IO/Extension Board.
 - USB-C or Micro-USB cable (capable of data transfer).
 - Power supply for the Waveshare board.
- **Software (Host Mac):**
 - usbboot (rpiboot) compiled and installed.
 - [Raspberry Pi Imager](#) installed.

Technical Workflow



Step-by-Step Procedure

1. Hardware Configuration (Boot Mode)

To allow the host computer to access the eMMC storage, the CM4 must be forced into USB boot mode.

1. Locate the **BOOT** or **nRPI_BOOT** switch/jumper on the Waveshare expansion board.

2. Set the switch to **ON** (or short the pins with a jumper cap).
3. Insert the CM4 module firmly into the expansion board slots.
4. Connect the data cable from your Mac to the **USB SLAVE** (or Type-C) port on the Waveshare board.
5. Connect the power supply to the Waveshare board and turn it on.

2. Mounting the eMMC File System

Because eMMC modules do not mount automatically like standard USB drives, you must use the rpiboot tool to initialize the connection.

1. Open the macOS **Terminal**.
2. Navigate to your local usbboot directory:

```
cd ~/usbboot
```

3. Execute the utility with root privileges:

```
sudo ./rpiboot
```

4. **Expected Output:** The terminal will display file transfer progress (loading bootcode.bin, msd.elf, etc.).
5. **System Notification:** macOS may display a warning stating *"The disk you inserted was not readable by this computer."* **Ignore this warning** and do not click initialize. The drive is now successfully exposed to the system.

3. Flashing the Operating System

1. Launch the **Raspberry Pi Imager** application.
2. Click **Choose OS** and select your desired operating system distribution.
3. Click **Choose Storage** and select the mounted CM4 eMMC drive (verify by matching the storage capacity, e.g., 8GB/16GB/32GB).
4. (Optional) Click the **Settings (Gear Icon)** to pre-configure hostnames, SSH access, user credentials, and Wi-Fi configurations.
5. Click **Write**. Enter your macOS admin password if prompted.
6. Wait for the writing and verification processes to complete successfully.

4. Post-Flashing & Normal Boot Target

Before the CM4 can boot independently, it must be taken out of programming mode.

1. Power off the Waveshare expansion board completely.
2. Disconnect the USB cable connecting the board to your Mac.
3. **CRITICAL:** Toggle the **BOOT** or **nRPI_BOOT** switch back to **OFF** (or remove the jumper cap). *Failure to do this will cause the board to hang on a blank screen waiting for a USB master link on subsequent boots.*
4. Connect your required peripherals (HDMI, Ethernet, Keyboard/Mouse).

5. Apply power to the board. The CM4 will now boot into the newly installed OS.

Troubleshooting

Symptom	Potential Cause	Resolution
rpiboot stuck on Waiting for BCM2835/6/7/2711...	Incorrect switch position or faulty cable.	Double-check that the BOOT switch is ON. Ensure the USB cable supports data transfer, not just power. Try a different USB port on the Mac.
Target drive not showing up in Pi Imager	rpiboot was not executed or failed.	Re-run <code>sudo ./rpiboot</code> in the terminal and ensure it finishes with a success message before opening the Imager.
Board won't boot into OS after flashing	BOOT jumper/switch is still enabled.	Turn off power, turn the BOOT switch to OFF , and power the board back on.